

Lead-Shielded Microfocus X-Ray NDT Cell

Authoritative Model Technical Manual & Diagnostic Reference | VAI-MAN-XR-001

NOTICE: SYNTHETIC PROTOTYPE TECHNICAL MANUAL

This document is artificially generated for the VectorAI demonstration and software development.

The specifications, thresholds, service-life values, maintenance intervals, and operating parameters are synthetic and must not be used for real industrial equipment operation or maintenance.

1. Machine Overview & Manufacturing Context

Equipment Model:	Lead-Shielded Microfocus X-Ray NDT Cell	Document ID:	VAI-MAN-XR-001
Machine Type Key:	x-ray-inspection	Cleanroom Bay / Stage:	Bay 5B: Lead-Shielded X-Ray
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Non-destructive testing (NDT) microfocus X-ray inspection station for detecting internal die-attach voids, wire bond sweeps, bridging, and solder ball voids within encapsulated IC packages.

Process Flow: A 130 kV open microfocus X-ray tube emits high-energy X-ray photons through encapsulated packages onto a high-resolution flat panel detector (FPD). A 5-axis motorized manipulator tilts and rotates the sample to generate 3D computed tomography (CT) slices.

2. Major Subsystems & Critical Components

Subsystem / Component	Primary Function	Key Parameters	Degradation Indicators
130 kV Microfocus Tube	Generates sub-micron focal spot X-ray beam.	High voltage (kV), target current (μA), target temp (°C).	Filament current degradation, target pitting, voltage arc.
Flat Panel Detector (FPD)	Converts X-ray photons into high-contrast digital radiograph.	Detector temperature (°C), pixel defect count, frame rate (fps).	Detector temperature rise, dead pixel cluster accumulation.
5-Axis Manipulator Stage	Positions and rotates IC packages in X/Y/Z/Tilt/Rotate axes.	Stage vibration (mm/s), rotation backlash (arcsec).	Manipulator mechanical chatter, positional drift.

3. Sensor Telemetry & Threshold Matrix

Threshold boundaries configured for real-time telemetry monitoring and automatic anomaly triggers:

Sensor Name & ID	Unit	Normal Range	Warning Range	Critical Range	Direction
Tube High Voltage tube_voltage	kV	120 – 135	135 – 148	148 – 160	Higher is Worse
Target Anode Temp tube_temp	°C	30 – 42	42 – 55	55 – 70	Higher is Worse

4. Deterministic RUL Model (Weighted Degradation)

Base Useful Life: 2,450 operating hours. **Model:** $RUL = BaseLife * (1 - (0.50 * wear_tube_temp + 0.30 * wear_voltage + 0.20 * wear_stage_vib))$

Parameter	Sensor ID	Unit	Weight	Healthy Limit	Critical Limit	Direction
Tube Temp	tube_temp	°C	0.50	35	60	HIGHER IS WORSE
Tube Voltage	tube_voltage	kV	0.30	130	155	HIGHER IS WORSE
Stage Vibration	vibration_stage	mm/s	0.20	0.25	0.8	HIGHER IS WORSE
TOTAL WEIGHT SUM	-	-	1.00 (100%)	-	-	VERIFIED OK

5. Troubleshooting & Anomaly Symptoms Matrix

Symptom & ID	Severity	Related Sensors	Possible Root Causes	Recommended Corrective Action
Radiographic Image Contrast Loss / Noise SYM-XR-001	HIGH	tube_voltage, tube_temp	• Tungsten target pitting • Chiller cooling restriction • FPD detector gain drift	Rotate target to fresh spot and verify chiller coolant flow.

6. Authoritative Diagnostic Knowledge Base (Failure Scenarios)

Scenario 1: X-Ray Target Anode Overheating & Voltage Instability (SCEN-XR-001)	Severity: WARNING
Telemetry Sensor Pattern:	tube_temp > 45°C, tube_voltage > 140 kV
Possible Root Causes:	Chiller heat exchanger fouling Filament emission drift
Recommended Corrective Action:	Flush tube cooling chiller and calibrate filament emission current.
Verification & Recovery Steps:	Measure cooling flow rate (> 3.2 L/min) > Verify image resolution with JIMA resolution test chart (< 2µm)